GPU-accelerated reconstruction of tiled spectral-domain OCT with removal of specular film reflections

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Abstract

Tiled spectral-domain optical coherence tomography (OCT) can image centimetre-scale tissue at micrometre resolution, but reconstruction is slow and samples mounted between transparent films show bright specular film reflections that mask the tissue surface. We present an open, GPU-accelerated reconstruction pipeline that reproduces the established myOCT tiled-scan reconstruction to within one output quantisation step while reducing the time for a 956-GB, 864-tile volume from an estimated 25.5 h to 18.4 min, limited only by storage bandwidth. We further introduce a per-A-line method that removes reflections of fluorinated ethylene propylene (FEP) films in the complex depth profile: each reflection is projected onto a rank-3 point-spread-function subspace learnt from the scan itself, shrunk to the local background level, so no dark band is left behind, and capped at the film's share of the energy estimated from its lateral phase coherence, so that tissue speckle is preserved (on real-data ground truth, the error on tissue without film is −24.2 dB relative to the tissue). Where the film is the only signal it falls from 16.7 to 2.4 dB above the background, deep tissue is unchanged, and the removed signal is exported for inspection. With the reflections gone, an automatically segmented tissue slab yields a tissue-only en-face projection: a single two-dimensional image of the section. The removal runs within the same GPU pass without increasing the full-volume reconstruction time (19.3 min versus 19.6 min), and a tile flat-field correction removes the brightness steps at the patch seams.

Introduction

Optical coherence tomography (OCT) measures depth-resolved backscattering by low-coherence interferometry [1,2]. In spectral-domain (Fourier-domain) OCT the depth profile of every lateral position (an A-line) is recovered from one spectrometer readout, which gives large gains in speed and sensitivity over time-domain systems [3–6]. To image specimens much larger than the field of view of a high-numerical-aperture objective, the sample is translated under the scan head and a mosaic of tiles is acquired at several focal depths; the tiles are then corrected for field curvature, weighted by their distance to focus and stitched into one volume. The open-source MATLAB toolbox myOCT [7] implements this tiled reconstruction for Thorlabs and Wasatch spectral-domain systems, and it is used to produce virtual-histology volumes.

Two problems limit this workflow at scale. First, reconstruction is slow: for the dataset studied here (12×9 lateral tiles at 8 focal depths, 864 tiles, 432,000 B-scans, 956 GB of raw spectra) the reference implementation needs 20.4 s per output plane, about 25.5 h for the 4500 planes, and its default parallel mode could not run on a 14-GB workstation. Graphics processing units (GPUs) are well established for real-time spectral-domain OCT processing of single B-scans [8–10], but a tiled, multi-depth reconstruction is dominated by data movement and by the bookkeeping of thousands of tiles, and to be useful it must reproduce the existing reference output exactly.

Second, many samples are mounted between thin transparent films—here fluorinated ethylene propylene (FEP)—to keep them flat and hydrated. The film surfaces are flat, smooth interfaces and act as mirrors. Each produces a specular reflection that is 10–30 dB brighter than the tissue and that lies, by construction, exactly at the tissue surface. In the stitched volume it appears as a bright band at $z \approx 0$ in every B-scan and, because the specular return depends on the angle between beam and film, as round, tile-periodic blobs in en-face planes and in projections. Lateral averaging, notch filtering or low-rank background subtraction either fail to suppress it or remove the tissue at the surface with it.

Here we describe `octrecon`, a Python reconstruction for NumPy/Numba (CPU) and CuPy/CUDA (GPU) that re-expresses each stage of the reference reconstruction as precomputed linear operators and fused kernels and streams the raw data tile by tile. We then introduce a physically motivated removal of film reflections that operates on the complex A-line before the magnitude is taken: the reflection is a single, shifted copy of the system point-spread function (PSF), whose complex depth profile spans a three-dimensional subspace, whereas tissue speckle spreads over all dimensions. Projecting onto that subspace and shrinking the projection down to the local background level removes the film without creating a dark band and leaves every sample outside a 17-sample window bit-identical. Because a single A-line cannot tell the film from a bright tissue speckle, the subtraction is capped at the film's share of the energy, estimated from the lateral phase coherence that only the film has, and the amount removed is kept smooth across neighbouring A-lines and B-scans so that no line artefacts are introduced. Finally, we correct the brightness falloff that makes the patch seams

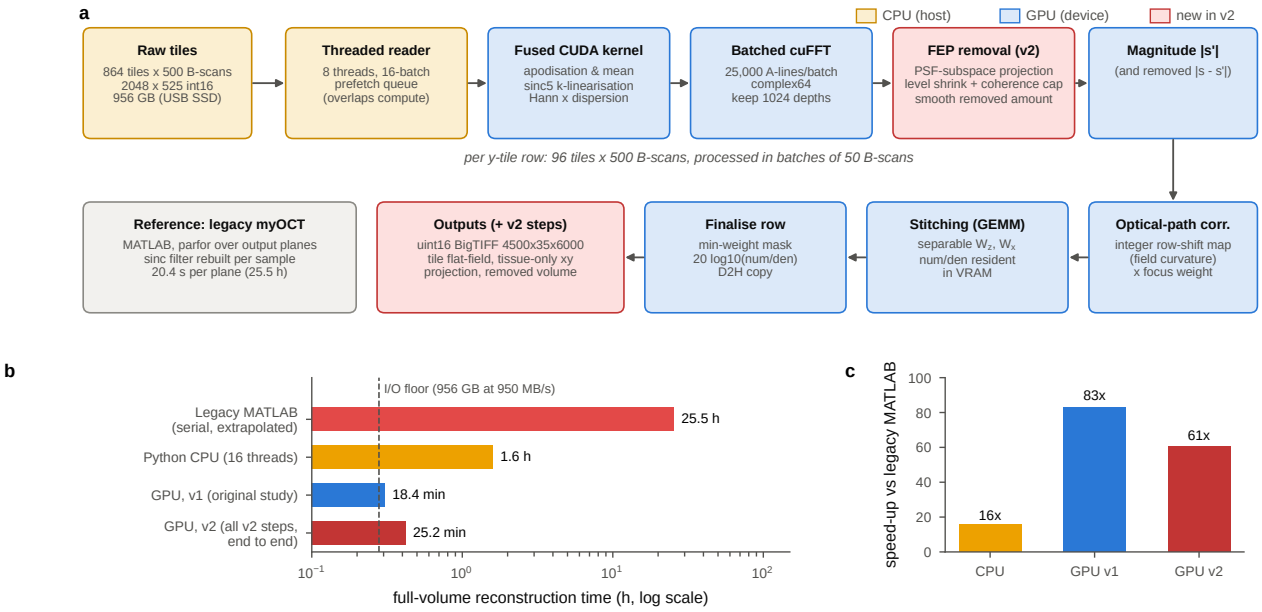


Fig. 1. Reconstruction pipeline and acceleration. **a**, Data flow for one row of tiles (96 tiles × 500 B-scans). Orange, host (CPU); blue, GPU; red, steps added in version 2. The FEP step (PSF-subspace projection, level shrinkage, coherence cap, smooth removed amount) operates on complex depth profiles before the magnitude; the stitched magnitude of the removed component is exported as a separate volume. After stitching, version 2 corrects the tile brightness falloff (flat-field) and writes the tissue-only projection. The grey box summarises the reference implementation (myOCT, MATLAB). **b**, Full-volume reconstruction time (log scale) for the dataset (864 tiles, 956 GB raw, $4500 \times 35 \times 6000$ output). Reference time is a measured plane (20.4 s) × 4500. CPU and GPU v1 values are measured end-to-end runs including the BigTIFF output (original study). GPU v2: end-to-end run with all version-2 steps (25.2 min: reconstruction 19.3 min, tile flat-field, projections and removed-signal volume). Dashed line, time to read 956 GB at 950 MB s^{-1}. **c**, Speed-up relative to the reference, end to end.

visible (a tile flat-field), segment the tissue in the cleaned volume and export a tissue-only en-face projection, a single two-dimensional image of the section to which neither air, film nor background contribute.

Results

A streaming GPU reconstruction that reproduces the reference output

The pipeline (Fig. 1a) follows the reference reconstruction step by step: subtraction of the apodisation spectrum, resampling to linear wavenumber with a five-lobe sinc kernel (sinc5), a Hann window with a quadratic dispersion phase, an inverse FFT, optical-path (field-curvature) correction, a Gaussian focus weight and a weighted-mean stitching into the output grid (Methods). Every operator that does not depend on the data—the sinc weights, the complex window, the curvature shift map and the interpolation matrices of the stitching—is computed once and kept on the device. The sinc5 resampling, which takes 77 % of the reference run time because the filter is rebuilt for each of the 2048 output samples of each B-scan, becomes a banded operator with at most 14 taps per sample and is fused with the apodisation and window into a single CUDA kernel; this alone made the spectral stage 6.2 times faster than an unfused GPU implementation. The 2D interpolations of each B-scan onto the full output plane become two small dense matrices per tile position, applied as batched matrix products to accumulators that stay in GPU memory for a whole row of tiles. Raw files are read tile-major by eight threads with 16 batches in flight, so disk reads overlap with computation.

On a laptop workstation (RTX 3080 Ti Laptop GPU, Ryzen 9 6900HX, 14 GB RAM, USB SSD) the full volume ($4500 \times 35 \times 6000$ output voxels at 2 µm) was reconstructed in 18.4 min end to end, 83 times faster than the reference extrapolated from a measured plane, and 5.3 times faster than the same code on all 16 CPU threads (1.6 h; Fig. 1b,c). The GPU waited for the disk 56 % of the time and read the raw data at an effective 877 MB s^{-1}, close to the drive limit: the reconstruction is input/output bound and the I/O floor for 956 GB is ≈17 min. Numerically, each stage agrees with the unmodified reference code to floating-point round-off; a stitched plane agrees to 9.3×10^{-5} dB in single precision, an order of magnitude below the output quantisation step (1.02×10^{-3} dB). Over all 945,000,000 voxels the GPU output lies within one quantisation step (1 LSB) of the existing reference volume, with identical masks of missing data (Extended Data Fig. 1); with the reference's two-step quantisation enabled it reproduces the reference file bit-exactly, which also identified the dispersion coefficient used for it ($\beta = 8.949 \times 10^7$ nm^2 rad^{-1}).

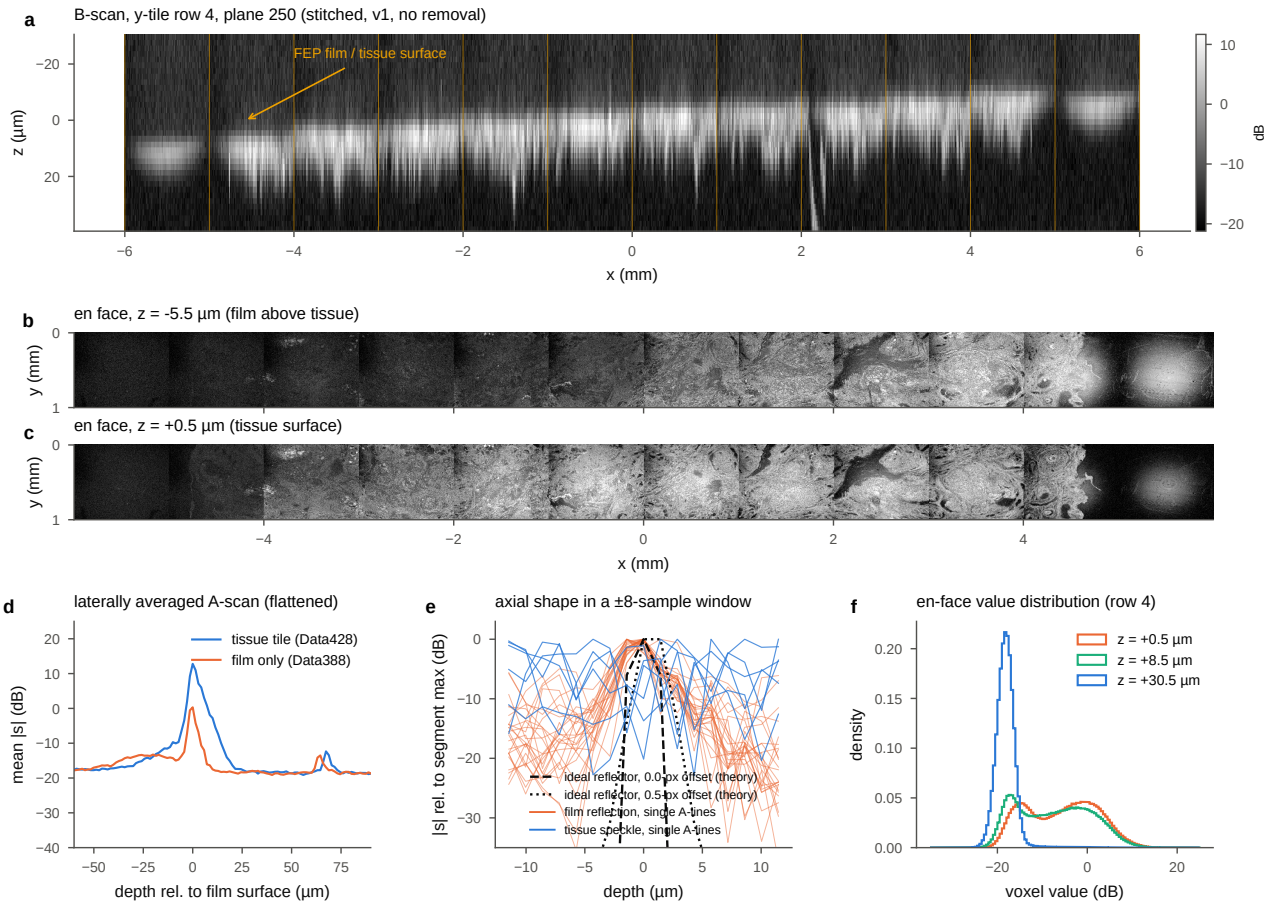


Fig. 2. Specular reflections of the FEP film. a, Stitched B-scan (row of tiles 4, plane 250; reconstruction without removal); vertical lines mark tile borders; vertical scale exaggerated. **b**,**c**, En-face planes of the same row of tiles (12×1 mm, y stretched $\approx 1.3\times$) at $z = -5.5\,\mu$m (film above the tissue) and $z = +0.5\,\mu$m (tissue surface). Note the round, tile-periodic blobs (clearest in the tissue-free tile on the right) and the tile-periodic brightness modulation. **d**, Laterally averaged, curvature-flattened magnitude of one B-scan of a tile with tissue (Data428) and of a tile with film only (Data388); 0 is the film surface, 1 pixel = 1.434 µm in tissue. **e**, Magnitude of 25 single A-lines of the film reflection (orange) and of six tissue A-lines (blue) in a ±8-sample window, each normalised to its maximum, with the theoretical response of an ideal reflector through the spectral window at integer and half-pixel position (black). **f**, Distribution of voxel values in en-face planes at the surface and in tissue (row of tiles 4).

Specular reflections of the FEP film

The tissue was mounted between FEP films. In stitched B-scans the film/tissue interface appears as a bright band at $z \approx 0$ (Fig. 2a). In en-face planes near the surface the film produces bright, round blobs whose centre repeats with the tile period (Fig. 2b,c; most visible in the tissue-free tile at $x \approx +5.5$ mm): the film is specular, so its return is strongest where beam and film are normal to each other, i.e. at the same position within every tile. The laterally averaged A-scan of a tissue tile shows a sharp peak 12.8 dB at the surface compared with a background of −13.9 dB (Fig. 2d), and the surface en-face plane has a bimodal value distribution with a bright mode that is absent in deep tissue (Fig. 2f).

Single A-lines reveal what distinguishes the film from tissue (Fig. 2e). Every film reflection has the same axial shape—the system PSF, a few pixels wide—and differs only by amplitude, phase and a sub-pixel shift; tissue speckle has a random axial structure. Across the field, the measured film amplitude fluctuates by about ±20 % from A-line to A-line (largely noise, see below) and its phase is random from B-scan to B-scan, which is why subtracting a laterally averaged complex reflection (in x or y) suppressed the film by only 10–17 dB in our tests; low-rank (SVD) background removal in a B-scan window failed because tissue at the surface is locally low-rank too, and notching the band destroys the tissue surface.

Removal by PSF-subspace projection and level shrinkage

A specular reflector at depth m contributes $a\,e^{i\varphi}p(z-m)$ to the complex A-line, where p is the complex PSF formed by the spectral window and the residual dispersion. For sub-pixel positions m these profiles span, within a 17-sample window ($\pm R$, $R = 8$), a subspace of very low dimension: one component captures 93.6 %

and three components >99.99 % of the energy of the theoretical family (Fig. 3a,c). Real reflections are not exactly theoretical, because of residual dispersion; we therefore learn the basis from the scan itself, from about 6,000 strong, phase-aligned film segments of 12 tiles spread over the grid (Methods). The learnt rank-3 basis (Fig. 3b) captures 96.8 % of the energy of a film reflection, against 85.5 % for the theoretical PSF (Fig. 3c,d; on the 2,082 segments of the two example tiles: 96.0 % versus 84.3 %).

For every A-line the method (i) locates each film surface, detected as a sharp peak of the laterally averaged, curvature-flattened profile and refined per A-line within ±4 samples followed by a 15-A-line lateral median, (ii) projects the complex 17-sample segment $\boldsymbol{s}$ onto the basis $\boldsymbol{U}$, $\boldsymbol{c} = \boldsymbol{U}^{\mathsf{H}}\boldsymbol{s}$, (iii) uses the fraction of energy captured, $f = \|\boldsymbol{c}\|^2 / \|\boldsymbol{s}\|^2$ ('specular-likeness'), to decide whether a film is present, and (iv) subtracts the projection, $\boldsymbol{s}' = \boldsymbol{s} - w\,g\,\boldsymbol{U}\boldsymbol{c}$, and (v) caps the subtraction at the film's share of the energy, estimated from the lateral coherence of the reflection (next sections). The weight w ramps from 0 to 1 between $f = 0.35$ and 0.60 (after a lateral median); film segments lie almost entirely above the ramp, whereas tissue and empty background have median f of 0.30 and 0.31 (Fig. 3d; Extended Data Fig. 2). Speckle has f above the white-noise value $r/(2R+1) = 0.18$ because it is axially band-limited by the same PSF; it is therefore excluded by the surface detection and by the ramp together, and processing is confined to detected surfaces.

The gain g is the key to avoiding artefacts. Tissue and noise also have energy along the PSF subspace, on average about $r\,P_{\mathrm{bg}}$ for a local background power P_{bg} per sample. Subtracting the whole projection ($g = 1$), or shrinking it with a Wiener-type rule $g = 1 - rP_{\mathrm{bg}}/\|\boldsymbol{c}\|^2$, removes this share as well and leaves a dark trench where the film was (Fig. 3e–g): on a film-only tile the surface fell from 0.3 dB to −25.2 dB (full projection) or −23.0 dB (Wiener), well below the −16.0 dB background. We instead shrink the coefficient to the background level, $g = \max\left(0,\, 1 - \sqrt{\kappa\, r\, P_{\mathrm{bg}}/\|\boldsymbol{c}\|^2}\right)$ with $\kappa = 0.7$, which leaves exactly $\kappa\, r P_{\mathrm{bg}}$ of energy in the subspace whatever the film amplitude (Fig. 3f). The surface then lands at −15.6 dB, at the background level, and on a tissue tile the 12.8 dB peak is reduced to −10.1 dB next to a local background of −13.9 dB (Fig. 3h). Samples outside the window are not modified.

Validation on synthetic ground truth

We injected PSF reflections with smooth sub-pixel tilt, a laterally coherent phase and an independent ±20 % amplitude jitter per A-line (the apparent fluctuation of the measured film) into simulated speckle tissue and evaluated the per-B-scan steps of the method (Extended Data Fig. 3; the shrinkage rules are compared without the coherence cap; with the cap the level rule gave −7.8, −6.9 and −6.6 dB; the smoothing of the removed amount across A-lines and B-scans is evaluated on real data below). Without removal the error in the 17-sample window was +1.7, +11.7 and +21.7 dB relative to the tissue for peaks 10, 20 and 30 dB above the tissue. With the rank-3 basis and the level rule it was −7.8, −6.9 and −6.6 dB, i.e. independent of the film amplitude; a rank-1 basis failed for strong reflections (+10.5 dB at 30 dB) because it cannot follow sub-pixel shifts. Full-projection and Wiener rules gave errors about 2 dB lower in tissue (−8.8 and −8.8 dB at 20 dB) but, on an empty background, the level rule kept the surface at the background level (+0.4 dB) and on real data only the level rule avoided the trench (Fig. 3g). Tissue without a reflection was left unchanged (the weight is zero), and reflections at or below the tissue level are deliberately left in place because they are not separable from speckle. The same criteria are enforced as unit tests of the software.

Protecting the tissue: the lateral-coherence cap

Per-A-line removal alone took too much tissue. A bright speckle grain at the tissue surface is itself shaped by the PSF and is captured by the basis almost as well as the film, so the decision and the shrinkage of a single A-line cannot tell them apart. In the reconstructed row of tiles, the uncapped method lowered the signal at the tissue surface by −11.3 dB at tile centres and by −7.7 dB at tile corners, where the film reflection is weak (Fig. 4c,d), and the removed-signal volume showed the tissue texture. What distinguishes the film is its lateral coherence: it is one continuous mirror, so its complex profile stays phase-consistent from A-line to A-line, whereas tissue speckle decorrelates beyond the lateral speckle size. We therefore estimate the film energy from the bias-corrected coherent average of the product of each segment with the segment 4 A-lines (8 µm) further, taken on the same rows, over 31 A-lines, and cap the subtraction at this estimate (Methods, equations (13),(14)).

We measured the effect against ground truth built entirely from real data: real tissue from just below the removal window (free of film signal) was placed under a real film reflection from a tile without tissue, at the film strength of the data and 10 dB weaker or stronger (Fig. 4a,b). Where there is no film, the uncapped method changed the tissue by −6.1 dB (correlation of the magnitudes with the truth 0.84); with the cap the change fell to −24.2 dB (1.00). Under a weak film (as at the tile corners) the error was −8.6 dB instead of −4.0 dB (0.93 vs 0.74), and at the film strength of the data −2.9 dB (0.74 vs 0.57). Film removal itself was unchanged: in the tiles without tissue the film stood 14.3 and 16.7 dB above the background before, and 1.9

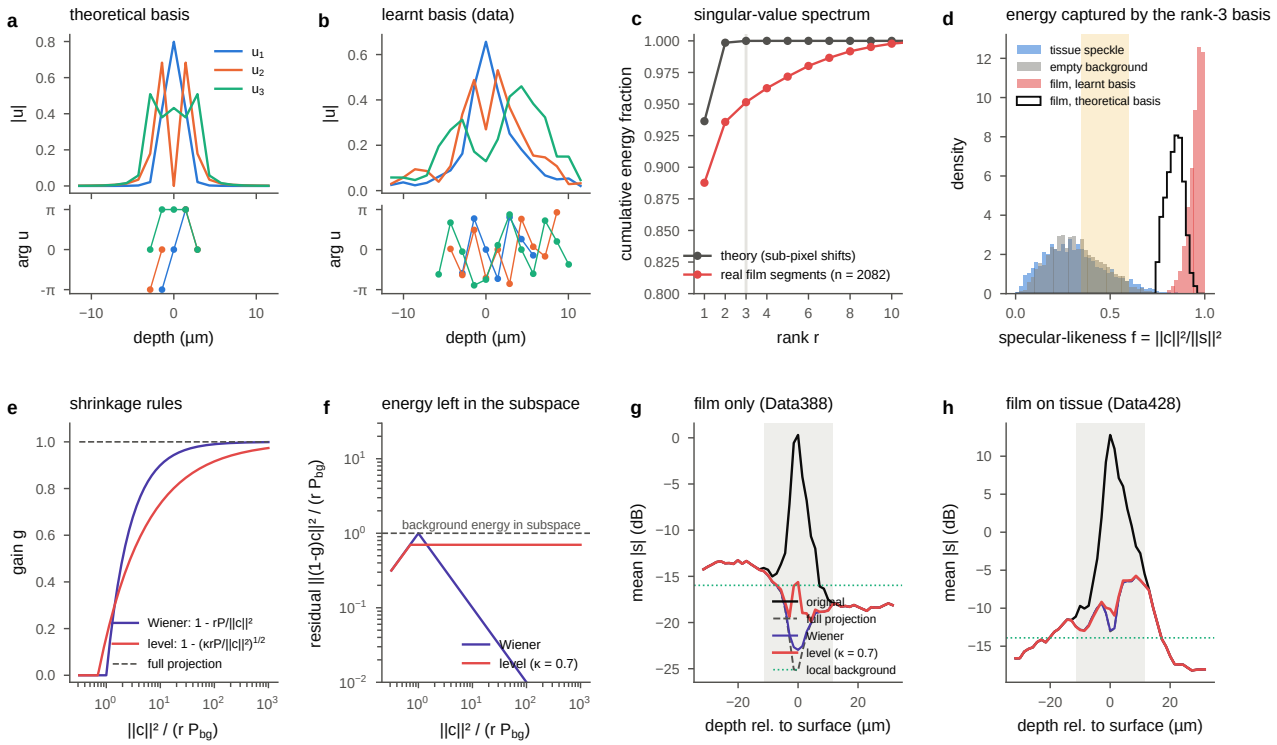


Fig. 3. Point-spread-function subspace and shrinkage rule. **a**, Rank-3 basis computed from the theoretical response of a reflector at 41 sub-pixel positions through the spectral window (magnitude, top; phase where $|u| > 0.1$, bottom). **b**, Rank-3 basis learnt from phase-aligned film segments of two tiles (Data388, Data428; 5 B-scans each). **c**, Cumulative energy fraction of the singular values for the theoretical family and for 2,082 real film segments. The grey line marks the rank used. **d**, Fraction of segment energy captured by the rank-3 basis for tissue speckle (Data428, 16–58 samples below the surface), empty background (Data388) and film segments (learnt and theoretical basis). Yellow, ramp of the soft weight w (0.35–0.60). **e**, Gain g versus the ratio of the projected energy to the expected background energy in the subspace, for the Wiener and the level rule ($\kappa = 0.7$). **f**, Energy left in the subspace after shrinkage, relative to the expected background energy; the level rule leaves κ of it for any film amplitude, the Wiener rule removes it as the film gets stronger. **g**,**h**, Laterally averaged, curvature-flattened profiles of one B-scan before and after removal for a film-only tile and a tissue tile. Grey band, $\pm R$ window; dotted, local background (10 rows on each side outside the window). Full projection and Wiener rule dig a trench below the background; the level rule does not.

and 2.4 dB after removal with the cap (1.9 and 2.0 dB without). In the tissue, the reduction at the surface fell to −2.0 dB at tile corners and −6.3 dB at tile centres, where it is dominated by the tile-centred film glow (Fig. 4c–e). For the strongest film (+10 dB) the residual is limited by the part of the reflection outside the rank-3 subspace (its wings), not by the cap.

Avoiding line artefacts

Every film decision is taken on one B-scan at a time. Taken independently, neighbouring B-scans (2 µm apart) received slightly different removal over runs of A-lines, which appeared in the en-face images as thin horizontal 'scraped' lines that version 1 does not have. Two properties of the film remove them. First, the field curvature rotates the film phase along the B-scan by a known amount (carrier phase times the optical-path shift), which we take out of the coherence products before averaging; without it the coherence estimate of a real film fluctuated between 0 and 1 along the B-scan. Second, the film is a smooth surface, so the amount removed must be smooth in x and y: position, weight and film energy are smoothed across five B-scans and the removed amplitude is replaced by a median-filtered, Gaussian-smoothed field, each A-line losing that amount along its own coefficient vector (Methods, equation (15)). On one row of tiles the energy of short horizontal dashes fell from −9.8 dB (per-B-scan removal) to −13.0 dB, against −14.2 dB for version 1, while the film in the tiles without tissue was still removed to the noise floor.

Real tissue volumes

Applied to a full row of tiles (500 output planes, 12 × 1 mm), removal eliminated the bright surface band and the tile-periodic blobs while preserving tissue structure (Fig. 5). Where the film is the only signal—the tiles without tissue—it fell from 14.3 and 16.7 dB above the background to 1.9 and 2.4 dB. Inside the tissue the changes are deliberately smaller, because the tissue under the film is kept: at the tissue surface ($z = +0.5$ µm)

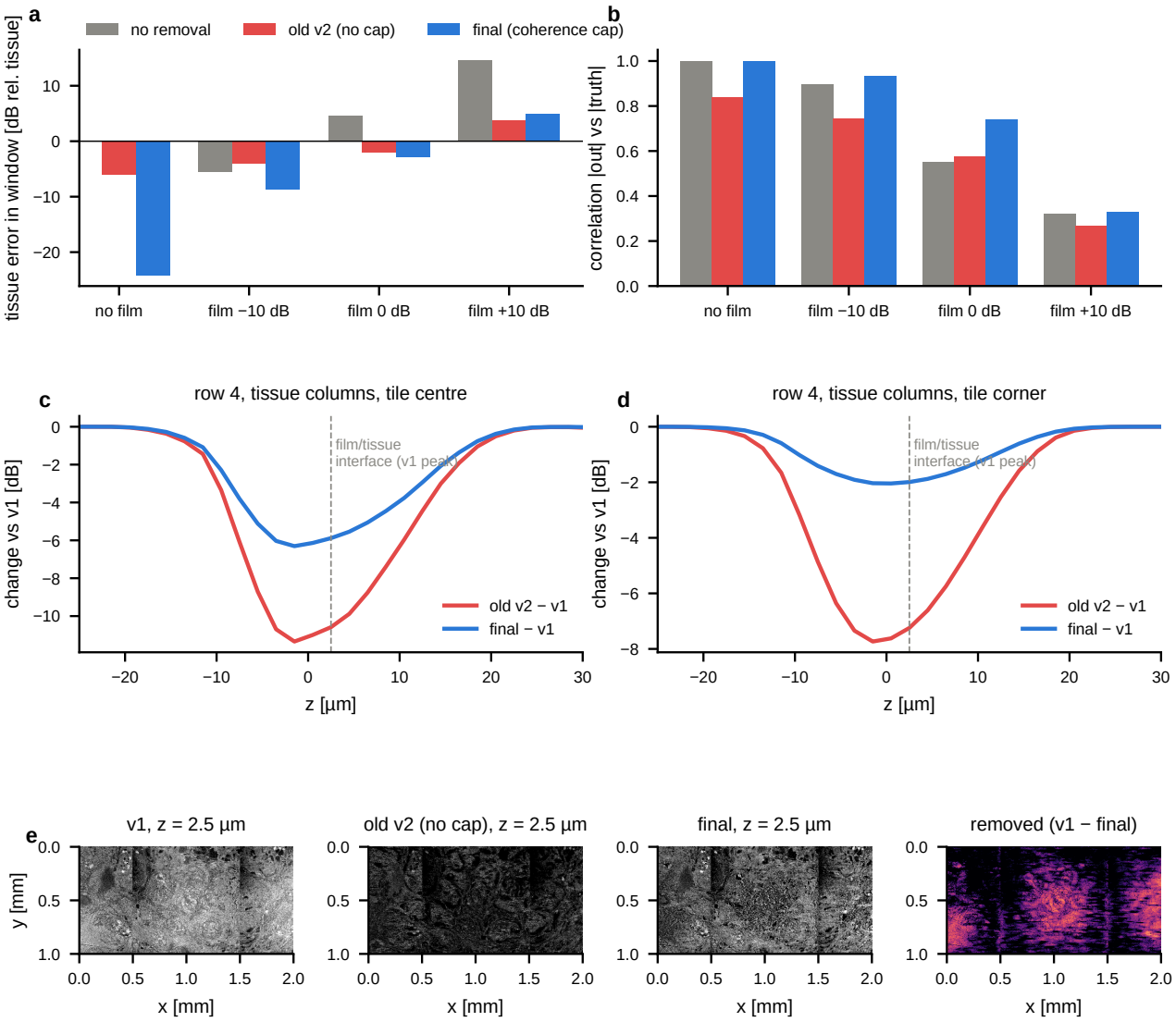


Fig. 4. Protecting the tissue with the lateral-coherence cap. **a**,**b**, Ground truth from real data: real tissue (from below the removal window, free of film signal) under a real film reflection from a tile without tissue, scaled to the strength found in the data (0 dB) and 10 dB weaker or stronger. **a**, Error of the output relative to the true tissue in the 9 rows under the film peak (dB relative to the tissue power; lower is better). **b**, Correlation of the output magnitudes with the true tissue magnitudes. Grey, no removal; red, per-A-line removal without the cap; blue, final method. **c**,**d**, Change of the stitched signal relative to version 1 (no removal) versus depth, averaged over the tissue columns of one row of tiles at the tile centres (strong film, $r < 100$ pixels from the tile centre) and corners (weak film, $r > 220$ pixels). Dashed, depth of the v1 maximum (the film/tissue interface). **e**, En-face planes 2.5 µm below the interface (2×1 mm, one tile and two halves): v1, without the cap, final method, and the amplitude removed by the final method (v1 minus final, same dB scale).

Table 1. En-face statistics before and after film-reflection removal (row of tiles 4, 500 × 6000 pixels per plane, values in dB as stored in the output volume).

z (µm)	region	median			99th percentile		
		before	after	Δ	before	after	Δ
−5.5	film above tissue	−10.7	−12.9	−2.2	7.9	5.1	−2.8
+0.5	tissue surface	−5.1	−9.2	−4.1	10.2	7.2	−2.9
+4.5	tissue	−5.1	−8.5	−3.3	10.7	7.8	−2.9
+8.5	tissue	−7.4	−10.7	−3.3	9.5	7.0	−2.5
+14.5	tissue	−14.7	−15.5	−0.9	6.0	4.8	−1.2
+30.5	deep tissue	−18.2	−18.2	0.0	−9.4	−9.5	−0.1

the median en-face value changed from −5.1 to −9.2 dB and the 99th percentile from 10.2 to 7.2 dB, and in the film plane above the tissue ($z = -5.5$ µm) the 99th percentile from 7.9 to 5.1 dB (Table 1). The effect decays with depth as the film window leaves the output plane, and deep tissue ($z = +30.5$ µm) is unchanged (median −18.2 versus −18.2 dB). Laterally, the y-averaged surface plane loses its tile-periodic arcs and the blob in the tissue-free tile disappears (Fig. 5d). The difference images (Fig. 5a, right) show that what is removed is concentrated in the two to three planes nearest the film and follows the illumination pattern of the tile rather than tissue morphology.

Tissue-only en-face projection and the removed-signal volume

For many uses a single two-dimensional image of the section is wanted. Version 2 therefore exports a calibrated *tissue-only* en-face projection: the tissue is segmented automatically in the reconstructed volume (lateral smoothing of the amplitude over 30 µm, Otsu threshold, hole filling; Methods), each column is reduced to the slab between its upper and lower tissue surface, and only voxels inside this slab contribute. The main image selects the depth per pixel: it averages the five strongest background-subtracted slices of the slab (P_{top}, about 10 µm), following the brightest tissue layer instead of diluting it with dark voxels; the mean amplitude (P_{mean}) and maximum (P_{max}) over the slab are also saved. Air, film, background and columns without tissue do not contribute and are left empty. The images are written with their physical pixel size (standard TIFF resolution and ImageJ calibration), the acquisition metadata and a display range set on the tissue. In addition the pipeline writes a 'removed-signal' volume: the stitched magnitude of exactly the component that was subtracted, $|\boldsymbol{s} - \boldsymbol{s}'|$, using the same curvature correction and stitching weights as the image. The removed volume makes the operation auditable: any tissue structure in it would indicate over-removal.

Film removal is a prerequisite for the tissue-only projection. With the same segmentation, the film reflections in tiles without tissue pass the threshold, so the 'tissue' footprint covers 82 % of the field without removal (round, tile-periodic blobs around the section, Fig. 6a) but 63 % with removal, which is the section itself (Fig. 6b; threshold −10.1 dB, median slab thickness 20 µm). Inside the tissue the bright surface band otherwise dominates every depth average. For one row of tiles, removal lowered the median of P_{mean} from −10.2 to −12.3 dB and the tile-periodic modulation (Methods) from 7.5 to 4.9 dB (Extended Data Fig. 4). For the full volume (Fig. 6) the tile-periodic modulation of P_{mean} is 6.7 dB without and 0.9 dB with film removal and the tile flat-field (next section), the median difference between the two projections is 2.6 dB, and the median removed amplitude is −20.5 dB (99th percentile −5.0 dB). The film removal, the tile flat-field and the v2 outputs can each be switched off; with all of them off the pipeline produces output identical to version 1.

Correcting the tile seams

Once the film was gone, the patch boundaries remained visible as a dark grid. The patches were acquired without overlap and neighbouring patches share no texture (Methods), so they cannot be registered or blended; the grid is a brightness falloff towards the patch edges that is the same in every patch and depends on depth (strongest at the tissue, absent in the background). We estimate this falloff $G(z, u, v)$ from the 53 patches that are fully covered by tissue and divide it out of the signal above the noise floor (Methods; Fig. 7c). The falloff reaches 24 dB at the patch corner where the fast scan starts (−8.3 dB along that edge at the tissue depth). In the tissue-only projection, the brightness step across the seams fell from 3.8 to 1.8 dB (seams between patch columns) and from 2.6 to 1.3 dB (between patch rows), against 0.7 dB for the same measure inside a patch, and the patch pattern folded over all patches fell from 5.4 to 1.5 dB (Fig. 7). What remains at the seams is the texture break between non-overlapping patches; acquiring with an overlap of about 10 % would allow registration and blending. The correction took 131 s for the full volume.

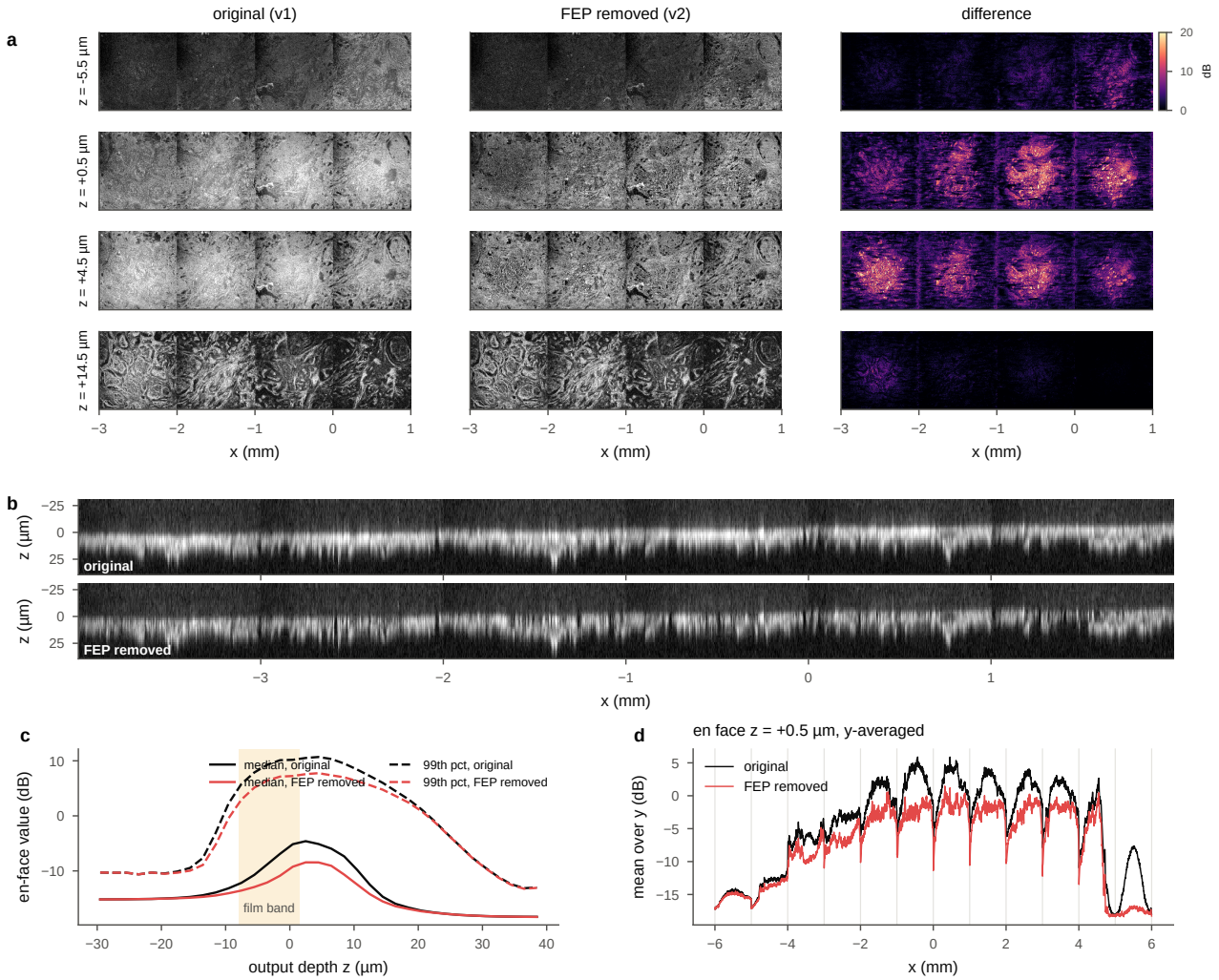


Fig. 5. Film-reflection removal on a stitched tissue volume. a, En-face planes at four depths of row of tiles 4 (4 mm of the 12-mm row shown) before (left) and after (middle) removal, displayed with the same grey scale (2nd–99.7th percentile of the original plane), and their difference in dB (right). **b**, B-scan (plane 250, 6 mm shown) before and after removal; vertical scale exaggerated. **c**, Median and 99th percentile of the en-face planes versus output depth. Yellow, depth range of the film reflection. **d**, Surface en-face plane averaged over y (amplitude average, in dB) across the whole row of tiles; grey lines mark tile borders.

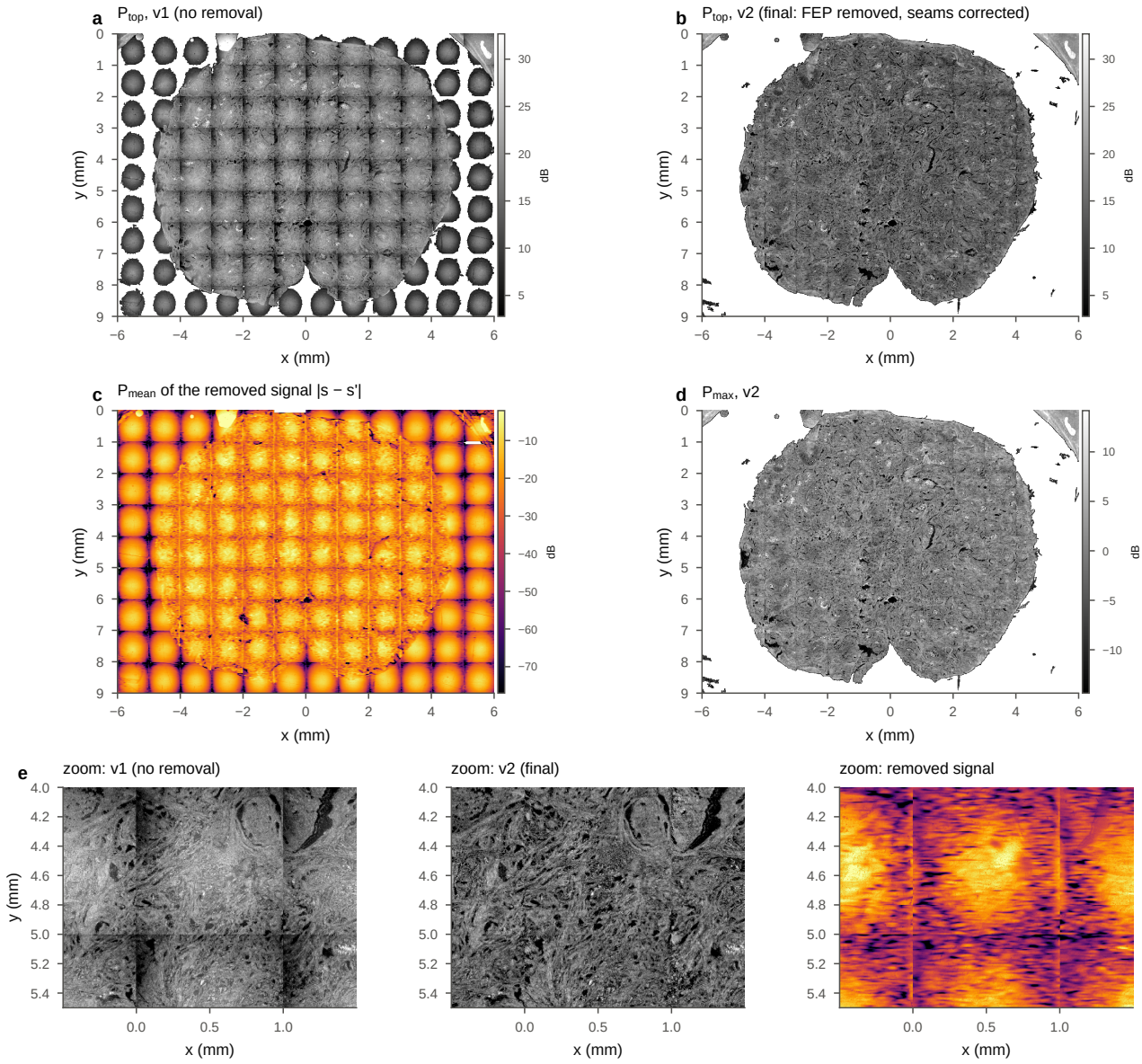


Fig. 6. En-face projections of the full volume and the removed signal. **a**,**b**, Tissue-only projection P_{top} (per pixel, mean of the five strongest background-subtracted slices of the segmented tissue slab; white/empty = no tissue) of the full 12×9 mm volume without (v1) and with all version-2 corrections (film removal, tile flat-field), on the same grey scale (4×4 block-averaged for display). Without removal the film reflections in tiles without tissue are segmented as tissue (round blobs) and brighten the projection inside the section. **c**, P_{mean} of the removed-signal volume $|\boldsymbol{s} - \boldsymbol{s}'|$, stitched with the same weights. **d**, Tissue-only maximum projection P_{max} of the v2 volume. **e**, Full-resolution zoom (2×1.5 mm across a tile corner) of **a**–**c**.

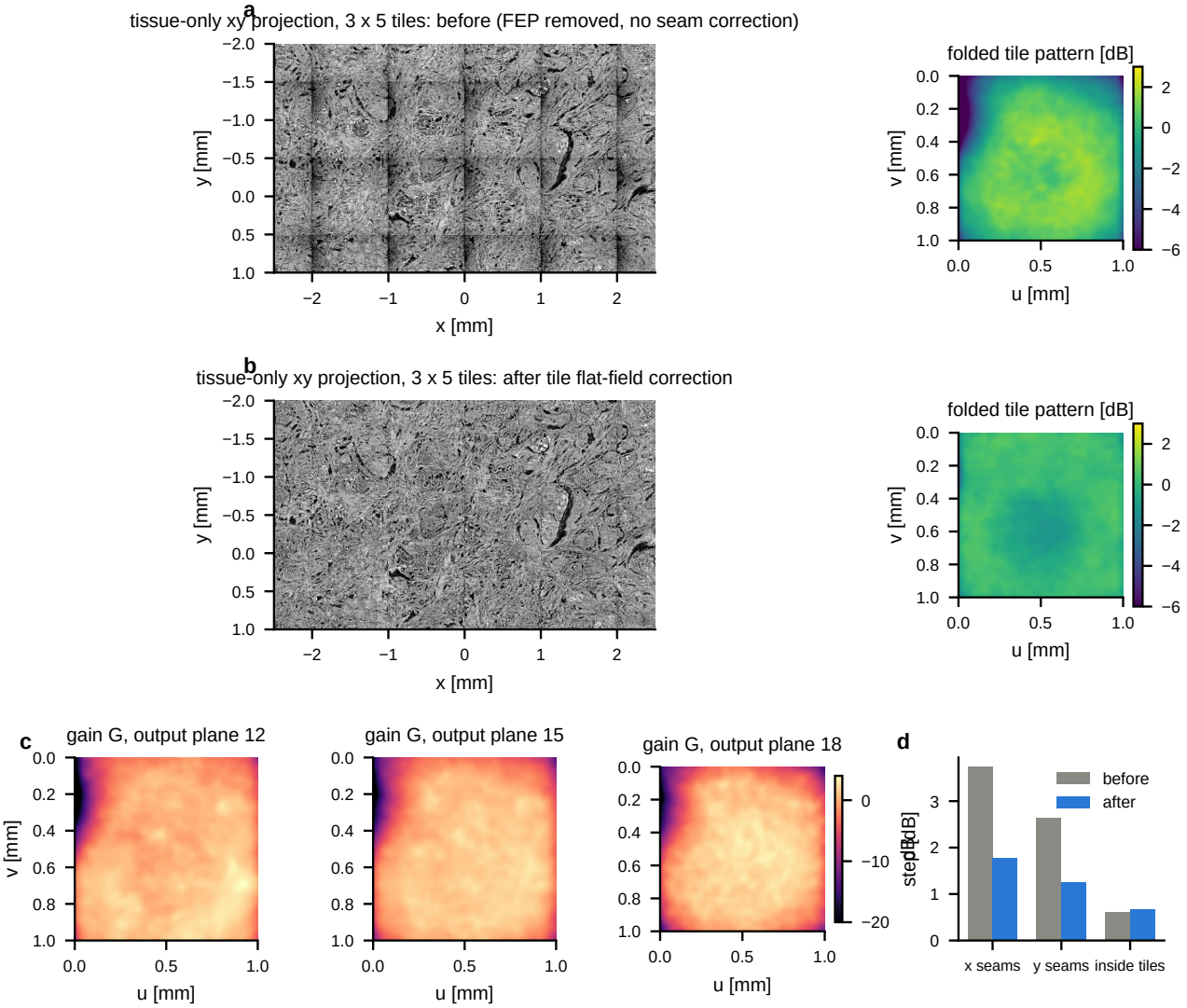


Fig. 7. Tile seam (flat-field) correction. **a**,**b**, Tissue-only projection over 3×5 patches before and after the correction, each with the patch pattern folded over all fully covered patches (right; dB). **c**, Estimated gain G of three output planes around the tissue depth (patch coordinates u, v; the fast scan starts at $u = 0$). **d**, Brightness step across the seams between patch columns (x) and rows (y), and the same measure inside the patches, before and after the correction.

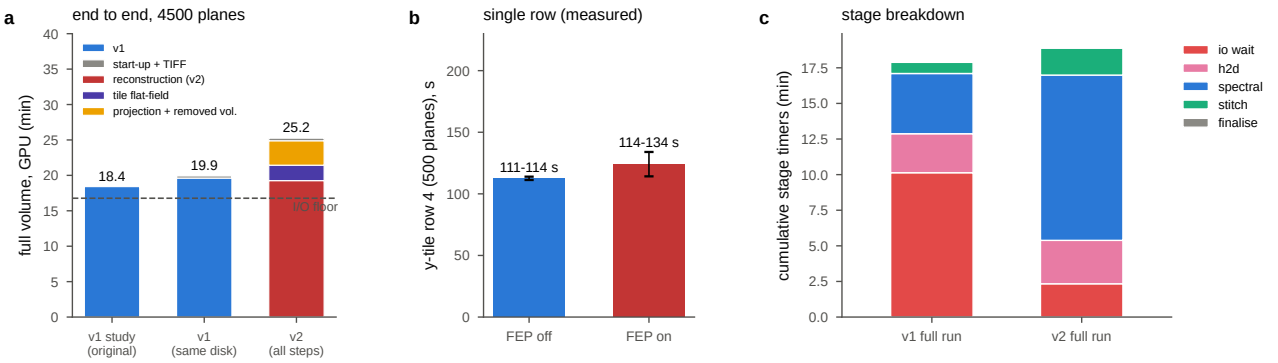


Fig. 8. Runtime. **a**, Full-volume GPU time, end to end (4500 planes): version 1 in the original study, version 1 on the same disk as version 2, and version 2 with all steps, split into reconstruction (including film removal), tile flat-field, projections with removed-signal volume, and start-up with TIFF writing. Dashed, I/O floor. **b**, Time for one row of tiles (500 planes, 96 tiles) with film removal off and on (range of two runs each, warm cache). **c**, Cumulative stage timers of the full runs (I/O wait, host-to-device copy, spectral processing, film removal, stitching, finalisation).

Runtime

Removal runs inside the same GPU pass, on the complex batch that is already on the device and only on the depth band that reaches the output; the lateral running medians of the small per-A-line maps (peak rows, weights, removed amount) also run on the GPU, which made them about 85 times faster than on the host. On one row of tiles in a warm cache the GPU time was 111–114 s without and 114–134 s with removal (two runs each; Fig. 8b). For the full volume, the reconstruction with all version-2 processing took 19.3 min (0.257 s per plane) against 19.6 min for version 1: the difference is within the run-to-run variation, because the full run is limited by reading the raw data from the external SSD (the pipeline waited 350 s for data without and 140 s with removal; the additional GPU work is hidden behind the I/O; Fig. 8c). Start-up including learning the basis took 11 s, the tile flat-field 131 s, and the tissue mask, projections and removed-signal volume another 3.5 min: 25.2 min end to end for version 2 against 19.9 min for version 1 on the same disk (Fig. 8a). End to end, version 2 is 61 times faster than the reference implementation, which has none of these steps.

Discussion

Two practical obstacles of large-area spectral-domain OCT were addressed in one open pipeline. The reconstruction of an 864-tile, 956-GB volume now takes minutes rather than a day on a laptop, and the result is the established reference reconstruction, not an approximation: the GPU output differs from it by at most one quantisation step and can reproduce it bit-exactly. Because the GPU is idle for more than half of the run, the remaining lever is storage bandwidth; the same code on faster storage, or with raw data compressed losslessly, would approach its compute limit of under 0.1 s per plane.

The film-reflection removal exploits the one property that separates a specular interface from tissue in a single A-line: its complex depth profile is a shifted copy of the PSF. The film is therefore removed within each A-line, along its own coefficient vector, which makes the method insensitive to the random B-scan-to-B-scan phase that defeated lateral averaging of the complex signal, and it keeps the signal outside a 17-sample window untouched. Only the *amount* removed is shared with the neighbours, as a smooth field across A-lines and B-scans; decided independently per B-scan it produced line artefacts. Learning the subspace from the scan absorbs residual dispersion and the actual spectral shape, which the theoretical PSF does not capture. The shrinkage to the background level is what makes the output visually and statistically plausible: a rule that is optimal for estimating the reflection (Wiener) is not optimal for estimating what lies beneath it, because the speckle component along the PSF subspace is removed as well. Keeping a fraction κ of the expected background energy is a simple, transparent compromise; the removed-signal volume documents exactly what was taken away.

The per-A-line model alone is, however, not enough to protect the tissue: a single A-line cannot tell the film from a bright speckle grain of the same shape, and the uncapped method removed several dB of tissue signal at the surface. The decisive physical difference is lateral phase coherence, which a film has and tissue speckle does not. Using it only to bound *how much* is removed—rather than to estimate the complex film signal by lateral averaging, which its random phase between B-scans defeats—combines the precision of the per-A-line model with the selectivity of the coherence: the film is removed as before, while on real-data ground truth the error on tissue without film is −24.2 dB relative to the tissue. The coherence products had to be corrected for the phase rotation that the field curvature imposes on the film; with that and a smooth removed

amount the coherence is a stable, artefact-free decision variable.

The method has limitations. Tissue that is itself specular and laterally coherent at the film interface—a flat, smooth tissue boundary pressed against the film, which reflects because of the refractive-index step—is indistinguishable from the film and is attenuated with it; part of the remaining reduction at tile centres is of this kind. Within ±1–2 samples of the surface, the image keeps the background intensity level but its axial speckle there is not the original. The decision thresholds were tuned on this dataset and may need adjustment for other films, objectives or spectrometers, although the basis itself is learnt automatically. Because the removed amount is smooth, a film far stronger than in this dataset (+10 dB) leaves a few dB more residual than a removal that follows every A-line; for this dataset the film in tiles without tissue was still removed to the noise floor. The tile flat-field removes the brightness steps at the patch seams but not the texture break between patches, which were acquired without overlap; an overlap of about 10 % would allow registration and blending. Finally, the flat-field assumes the same falloff in every patch, as for one scan pattern and objective, and is capped where the falloff is extreme.

Methods

Imaging system and dataset

Data were acquired with a Thorlabs GAN632 spectral-domain OCT system (OCITY software) with an Olympus 20× objective. [Sample description and preparation.] The tissue was mounted between FEP films. The tiled scan comprised 12 $(x) \times 9$ (y) lateral tiles of 1×1 mm at 8 focal depths (864 tiles). Each tile holds 500 B-scans; each B-scan file contains 525 A-lines (25 apodisation A-lines and 500 image A-lines) of $N = 2048$ spectral pixels stored as int16, 956 GB in total. The output volume is 4500 $(y) \times 35$ $(z) \times 6000$ (x) voxels at 2 µm isotropic spacing, covering $z \in [-29.5,\, 38.5]$ µm relative to the tissue interface. All parameters are listed in Table 2; they are read from the scan metadata (`ScanInfo.json`, `Header.xml`, chirp file) or from the probe preset, except where stated.

Spectral-domain signal model and per-A-line processing

For a lateral position $\boldsymbol{x} = (x, y)$ the spectrometer records, for pixel $j = 0, \ldots, N-1$,

$$I_j(\boldsymbol{x}) = S(k_j)\Big[R_\mathrm{r} + \sum_n R_n + 2\sum_n \sqrt{R_\mathrm{r} R_n}\cos\big(2k_j n_\mathrm{g} z_n + \phi_n(k_j)\big) + \ldots\Big], \tag{1}$$

where S is the source spectrum, R_r and R_n the reference and sample reflectivities, z_n the depth of scatterer n, n_g the group index and ϕ_n a dispersion phase; the dots denote mutual interference terms of the sample [2,4]. The wavelength of pixel j follows from the chirp c_j of the spectrometer calibration, $\lambda_j = \big[c_j(\lambda_\mathrm{max}^{-1} - \lambda_\mathrm{min}^{-1})/(N-1) + \lambda_\mathrm{min}^{-1}\big]^{-1}$, with $[\lambda_\mathrm{min}, \lambda_\mathrm{max}] = [796, 1010]$ nm, and $k_j = 2\pi/\lambda_j$.

Apodisation and background subtraction. The mean of the $A = 25$ apodisation A-lines of the same B-scan file (recorded with the beam outside the sample) is subtracted from every image A-line, $I'_j(\boldsymbol{x}) = I_j(\boldsymbol{x}) - \frac{1}{A}\sum_{a=1}^{A} I_j(a)$.

k-linearisation. The spectrum is resampled onto a uniform wavenumber grid $\tilde{k}_i$, $i = 0, \ldots, N-1$, running linearly from $\max k$ to $\min k$. With fractional source positions $n_j = (k_j - \min k)/(\max k - \min k)\,(N-1)$ and target positions $q_i = N - 1 - i$, the sinc5 interpolation is

$$\tilde{I}_i = \sum_{j:\,|q_i - n_j| < 5} \mathrm{sinc}(q_i - n_j)\,\big(I'_j - \bar{I}'\big) + \bar{I}', \tag{2}$$

where $\bar{I}'$ is the spectral mean of the A-line. The weights depend only on the chirp; they are precomputed as a banded operator with $T \le 14$ non-zero taps per output sample.

Window, dispersion compensation and Fourier transform. The resampled spectrum is multiplied by the complex window

$$W_i = \frac{h_i}{\mathrm{rms}(h)}\exp\big(-i\beta(\tilde{k}_i - \langle\tilde{k}\rangle)^2\big), \tag{3}$$

with h a Hann window of length N, $\tilde{k}$ in rad nm^{-1} and the quadratic dispersion coefficient $\beta = 8.949 \times 10^7$ nm^2 rad^{-1}, which corrects the dominant second-order dispersion mismatch between the arms [11]. The

complex A-line is

$$s_m(\boldsymbol{x}) = \frac{1}{N}\sum_{i=0}^{N-1} W_i\,\tilde{I}_i(\boldsymbol{x})\,e^{2\pi i\,im/N}, \qquad m = 0,\dots,N/2-1, \tag{4}$$

sampled at depths $z_m = m\,\Delta z$ with $\Delta z \approx \bar{\lambda}^2/(2n\,\Delta\lambda) = 1.434\,\mu\text{m}$ in tissue ($n = 1.33$; $\bar{\lambda}$ and $\Delta\lambda$ the mean and span of the wavelengths). The image magnitude is $A_m(\boldsymbol{x}) = |s_m(\boldsymbol{x})|$; in version 2 the film removal (below) is applied to s_m first.

Optical-path (field-curvature) correction. The scan geometry adds a path-length offset $P(x,y) = p_1x + p_2y + p_3x^2 + p_4y^2 + p_5xy$ (µm; tile-local x, y in µm; coefficients of the probe in Table 2). Nearest-neighbour resampling at $z_m + P$ on the uniform depth grid is an integer row shift $\delta(\boldsymbol{x}) = \lfloor P(\boldsymbol{x})/\Delta z + \frac{1}{2}\rfloor$,

$$A^{\mathrm{c}}_m(\boldsymbol{x}) = A_{m+\delta(\boldsymbol{x})}(\boldsymbol{x}), \qquad v_m(\boldsymbol{x}) = \big[\,0 \le m + P(\boldsymbol{x})/\Delta z \le N/2-1\,\big], \tag{5}$$

with invalid samples ($v = 0$) set to zero. Across a tile δ spans up to 37 rows ($\approx 53\,\mu\text{m}$).

Focus weighting and stitching. Each tile is placed in the output frame at its lateral centre and focal depth, with its depth axis re-referenced so that $z = 0$ lies at the focus pixel $m_{\mathrm{f}} = 433$. A focus weight

$$F_m = \exp\!\left(-\frac{(m-m_{\mathrm{f}})^2}{(2\sigma)^2}\right) + e^{-4.5}, \qquad \sigma = 10\ \text{px}, \tag{6}$$

favours in-focus data (effective Gaussian standard deviation $\sqrt{2}\sigma$; the constant floor keeps out-of-focus data where no other tile contributes). Linear interpolation from tile coordinates to the output grid is separable; for tile t it is written with an output-depth matrix $\boldsymbol{W}_z^{(t)}$ and an output-column matrix $\boldsymbol{W}_x^{(t)}$ (zero outside the tile). For every output plane y,

$$\mathrm{num}(z,x) = \sum_t \Big[\boldsymbol{W}_z^{(t)}\big(\boldsymbol{A}^{\mathrm{c}} \odot \boldsymbol{v} \odot \boldsymbol{F}\big)^{(t)}\boldsymbol{W}_x^{(t)\mathsf{T}}\Big]_{zx}, \quad \mathrm{den}(z,x) = \sum_t \Big[\boldsymbol{W}_z^{(t)}\big(\boldsymbol{v} \odot \boldsymbol{F}\big)^{(t)}\boldsymbol{W}_x^{(t)\mathsf{T}}\Big]_{zx}, \tag{7}$$

summing over the 96 tiles that contribute to the plane. Voxels with $\mathrm{den} < e^{-4.5}(1-8\epsilon)$ (ϵ, machine precision of the accumulator; the factor protects voxels that sit exactly on the threshold) are set to NaN, and the output is

$$I(x,y,z) = 20\log_{10}\big(\mathrm{num}(z,x)/\mathrm{den}(z,x)\big). \tag{8}$$

Quantisation. The volume is written as a uint16 BigTIFF with the global range $[c_1, c_2] = [\min I, \max I]$, $Q = \mathrm{round}\,\big((I-c_1)/(c_2-c_1)\cdot 65534\big) + 1$ and $Q = 0$ for NaN; $1\,\mathrm{LSB} = (c_2-c_1)/65534 \approx 1.02\times10^{-3}\,\mathrm{dB}$. The reference implementation quantises each plane to its own range first and re-quantises to the global range; this two-step rounding is available as an option and was used for the bit-exact comparison. Metadata are stored as JSON in the first page and as a sidecar file.

Model of the film reflection

Around a film surface at (fractional) depth $\mu(\boldsymbol{x})$ the complex A-line is

$$s_m(\boldsymbol{x}) = a(\boldsymbol{x})\,e^{i\varphi(\boldsymbol{x})}\,p\big(m-\mu(\boldsymbol{x})\big) + t_m(\boldsymbol{x}) + \eta_m(\boldsymbol{x}), \tag{9}$$

with p the complex system PSF, a and φ the amplitude and phase of the reflection, t tissue speckle and η noise. For an ideal reflector and perfect dispersion compensation, Eq. (4) gives $p(m-\mu) = \frac{1}{N}\sum_i |W_i|\,e^{2\pi i\,i(m-\mu)/N}$. In a window of $2R+1$ samples centred on the nearest integer m_0 of μ, the profiles for offsets $d = \mu - m_0 \in [-\frac{1}{2}, \frac{1}{2}]$ form a smooth one-parameter family, which is well approximated by a subspace of dimension $r \ll 2R+1$ (Fig. 3c). Tissue speckle[12], by contrast, is a random superposition of PSFs at all depths and distributes its energy over the whole window; its expected energy within an r-dimensional subspace is about $r\,P_{\mathrm{bg}}$ for a local power P_{bg} per sample (more for axially correlated speckle; Fig. 3d). The measured film amplitude fluctuates by about ±20 % between neighbouring A-lines (in tiles without tissue the noise alone accounts for about 15 %, and a laterally smooth removal takes the film down to the noise floor, so the true amplitude varies smoothly), and φ is random between B-scans, so the complex reflection cannot be predicted from neighbours; it is estimated within each A-line, and only its amount is shared between neighbours (step 7).

PSF basis

Theoretical basis. For 41 offsets d uniformly spaced in $[-\frac{1}{2}, \frac{1}{2}]$ we compute

$$a_d(m) = \mathrm{IFFT}_n\big\{\,|W_n|\; e^{-2\pi i\, n(m_0+d)/N}\big\}(m), \qquad m_0 = N/4, \tag{10}$$

take the samples $m_0 - R, \ldots, m_0 + R$, normalise each to unit norm, and keep the first r left singular vectors of the $(2R+1) \times 41$ matrix as $\boldsymbol{U}_{\mathrm{th}}$.

Learnt basis. Before processing, 12 tiles are selected on a $4\,(x) \times 3\,(y)$ grid at the focal depth closest to $z = 0$, and five B-scans near the centre of each tile (every 4th of 20) are reconstructed to complex A-lines. Surfaces are detected as described below; for each surface and A-line the peak is located, the $(2R+1)$-sample segment $\boldsymbol{s}_k$ around it extracted and kept if it is clearly specular ($\big\|\boldsymbol{U}_{\mathrm{th}}^{\mathsf{H}}\boldsymbol{s}_k\big\|^2 / \|\boldsymbol{s}_k\|^2 \geq 0.75$) and at least as bright as the median segment. The kept segments are phase-aligned and normalised, $\hat{\boldsymbol{s}}_k = \boldsymbol{s}_k\, e^{-i \arg s_{k,R}} / \|\boldsymbol{s}_k\|$, at most 6000 are drawn at random, and the basis $\boldsymbol{U}$ is formed by the first r left singular vectors of $[\hat{\boldsymbol{s}}_1, \ldots, \hat{\boldsymbol{s}}_K]$. If fewer than 50 segments are found, $\boldsymbol{U}_{\mathrm{th}}$ is used. A basis can also be supplied as a file.

Film-reflection removal

The removal operates per tile on each batch of complex B-scans, before the magnitude, and only on the depth band $[r_0, r_1]$ that reaches the output after curvature correction (extended by $R + S$ samples).

(1) *Surface detection.* The curvature-flattened magnitude $\big|s_{r+\delta(\boldsymbol{x})}(\boldsymbol{x})\big|$ is averaged over the B-scans of the batch and the central 70 % of the A-lines, giving a profile $L(r)$ in dB. A row r is a surface candidate if $L(r) = \max_{|r'-r|\leq 3} L(r')$ and $L(r) - \mathrm{med}_{51} L(r) \geq 6\,\mathrm{dB}$, where med_{51} is a 51-row running median (the local baseline). Candidates are accepted strongest first if they are more than R rows apart, up to four surfaces (top and bottom film surfaces and their back sides).

(2) *Per-A-line position.* For surface r the expected native row is $r + \delta(\boldsymbol{x})$; the peak $\hat{m}(\boldsymbol{x}) = \arg\max_{|m-r-\delta(\boldsymbol{x})|\leq S} |s_m(\boldsymbol{x})|$ ($S = 4$) is smoothed with a 15-A-line lateral median and rounded, $\tilde{m}(\boldsymbol{x})$, so that a bright speckle grain cannot capture it.

(3) *Projection.* With $\boldsymbol{s}(\boldsymbol{x}) = \big(s_{\tilde{m}-R}, \ldots, s_{\tilde{m}+R}\big)^{\mathsf{T}}$, $\boldsymbol{c} = \boldsymbol{U}^{\mathsf{H}}\boldsymbol{s}$ and the specular-likeness $f = \|\boldsymbol{c}\|^2 / \|\boldsymbol{s}\|^2$.

(4) *Soft decision.* $w = \mathrm{clip}\big((\mathrm{med}_{15} f - f_{\mathrm{lo}})/(f_{\mathrm{hi}} - f_{\mathrm{lo}}), 0, 1\big)$, $f_{\mathrm{lo}} = 0.35$, $f_{\mathrm{hi}} = 0.60$, with med_{15} the lateral median over 15 A-lines.

(5) *Shrinkage to the background level.* With the local background power

$$P_{\mathrm{bg}} = \frac{1}{2B} \sum_{b=1}^{B} \Big(|s_{\tilde{m}-R-b}|^2 + |s_{\tilde{m}+R+b}|^2 \Big), \qquad B = 10, \tag{11}$$

the gain and the corrected segment are

$$g = \max\left(0,\; 1 - \sqrt{\frac{\kappa\, r\, P_{\mathrm{bg}}}{\|\boldsymbol{c}\|^2}}\right), \qquad \boldsymbol{s}' = \boldsymbol{s} - w\, g\, \boldsymbol{U}\boldsymbol{c}, \qquad \kappa = 0.7. \tag{12}$$

For $g > 0$ the energy left along the subspace is $\|(1-g)\boldsymbol{c}\|^2 = \kappa\, r\, P_{\mathrm{bg}}$, i.e. a fraction κ of what the background is expected to have there, independently of the film amplitude (a shrinkage in the spirit of soft thresholding[13], applied to the subspace energy). The Wiener-type rule $g_{\mathrm{W}} = \max(0, 1 - r P_{\mathrm{bg}} / \|\boldsymbol{c}\|^2)$ leaves $(r P_{\mathrm{bg}})^2 / \|\boldsymbol{c}\|^2$, which vanishes for strong reflections and produced a dark band on the data (film-only tile: $-23.0\,\mathrm{dB}$ at the surface against a $-16.0\,\mathrm{dB}$ background); $\kappa = 0$ subtracts the full projection.

(6) *Lateral-coherence cap (tissue protection).* Steps (3)–(5) act on single A-lines and cannot distinguish a film reflection from a bright, PSF-shaped tissue speckle grain at the surface. The film is a continuous mirror, so its complex profile is phase-consistent between A-lines, whereas tissue speckle decorrelates beyond the lateral speckle size. With $\boldsymbol{s}_L(\boldsymbol{x})$ the segment of the A-line L positions further along the B-scan, taken on the *same* rows $\tilde{m}(\boldsymbol{x}) \pm R$,

$$q(\boldsymbol{x}) = \boldsymbol{s}_L(\boldsymbol{x})^{\mathsf{H}} \boldsymbol{s}(\boldsymbol{x}), \qquad \hat{E}_{\mathrm{f}}(\boldsymbol{x}) = \sqrt{\max\Big(0,\; \big|\langle q \rangle_W\big|^2 - \tfrac{1}{W}\big\langle\, |q|^2 \,\big\rangle_W\Big)}, \tag{13}$$

where $\langle\cdot\rangle_W$ is a running mean over $W = 31$ A-lines and the second term removes the bias of incoherent products ($\mathbb{E}\,|\langle q \rangle_W|^2 = \langle |q|^2 \rangle_W / W$ for random phases). For a film, $\hat{E}_{\mathrm{f}} \approx \|\boldsymbol{s}\|^2$; for tissue alone, $\hat{E}_{\mathrm{f}} \approx 0$. The gain is capped at the film's share,

$$g \;\leftarrow\; g \cdot \min\Big(1,\; (1+\mu)\sqrt{\hat{E}_{\mathrm{f}} / \|\boldsymbol{c}\|^2}\Big), \qquad L = 4\;(8\,\mu\mathrm{m}), \quad \mu = 1, \tag{14}$$

with the margin μ absorbing amplitude fluctuations of the film and the remaining decorrelation of q (below). The lag exceeds the lateral speckle size (lags of 1–2 A-lines, where adjacent A-lines share speckle, removed more tissue). Before averaging, q is demodulated for the phase rotation that the field curvature imposes on the film: a reflector at fractional depth m (rows) has the carrier phase $-2\pi c_0 m$ with $c_0 = k_0/(N\Delta k) \approx 4.72$, and the film depth of A-line $\boldsymbol{x}$ is $r + \delta(\boldsymbol{x})$ with the known optical-path shift δ, so $q \leftarrow q \exp\{\pm i\, 2\pi c_0[\delta(\boldsymbol{x}+L) - \delta(\boldsymbol{x})]\}$ (sign chosen once per scan as the one that maximises $|\langle q\rangle_W|$).

(7) *A smooth removed amount (no line artefacts).* Steps (2)–(6) are evaluated per B-scan. Decided independently, neighbouring B-scans (2 µm apart) received slightly different removal over runs of A-lines, which appeared as thin lines in en-face images. Because the film is a smooth surface, the peak position, the weight w and $\hat{E}_\mathrm{f}$ are smoothed across 5 B-scans, and the removed amplitude $a = w\, g\, \|\boldsymbol{c}\|$ is replaced by

$$\bar{a} = \mathcal{G}[\mathrm{med}_{5\times 31}\, a]\,, \qquad \boldsymbol{s}' = \boldsymbol{s} - \min(1, \bar{a}/\,\|\boldsymbol{c}\|)\, \boldsymbol{U}\boldsymbol{c}, \tag{15}$$

a median over 5 B-scans × 31 A-lines followed by a Gaussian $\mathcal{G}$ ($\sigma = 1.7$ B-scans, 5 A-lines): every A-line loses a laterally smooth amount, along its own coefficient vector.

Samples outside $\tilde{m} \pm R$ are bit-identical to the input; inside the window only the r components along $\boldsymbol{U}$ change. The heavy operations (gathers, projections, updates) run on the device in the batch; the running medians and coherence averages act on small ($B \times n_x$) maps (on the GPU with the CUDA backend).

Removed-signal volume. The subtracted component $\Delta\boldsymbol{s} = \boldsymbol{s} - \boldsymbol{s}' = \min(1, \bar{a}/\,\|\boldsymbol{c}\|)\, \boldsymbol{U}\boldsymbol{c}$ (summed over surfaces) is kept as a complex array; its magnitude $|\Delta s_m|$ is curvature-corrected, weighted and stitched with exactly the same $\boldsymbol{v}$, $\boldsymbol{F}$, $\boldsymbol{W}_z$, $\boldsymbol{W}_x$ and denominator as the image, and written as a second volume $I_\mathrm{rem} = 20\log_{10}(\mathrm{num}_\mathrm{rem}/\mathrm{den})$.

Tissue segmentation and tissue-only en-face projections

The stitched dB volume $I(x, y, z)$ is processed in chunks of 256 y-planes directly from the memory-mapped result (three streaming passes), so the step works for volumes larger than the main memory. *(i) Lateral smoothing.* Per depth plane the amplitude $A = 10^{I/20}$ is averaged over an $L \times L$ window in (x, y) ($L = 30$ µm $= 15$ pixels) by normalised convolution, ignoring voxels without data:

$$S(x, y, z) = 20\log_{10} \frac{(⊮_L * A\,\chi)(x, y, z)}{(⊮_L * \chi)(x, y, z)}, \tag{16}$$

with χ the indicator of finite voxels and $⊮_L$ the box kernel. Averaging over the speckle separates the volume-scattering tissue from the noise floor. *(ii) Threshold.* The threshold τ is Otsu's threshold of the histogram of all S values (0.05 dB bins), i.e. the value maximising the between-class variance $\omega_0(\tau)\,\omega_1(\tau)\,[\mu_0(\tau) - \mu_1(\tau)]^2$; the voxel mask is $M = S > \tau$. *(iii) Footprint.* A column is tissue if it contains at least two masked voxels; connected components smaller than 0.005 mm^2 are removed, and enclosed non-tissue regions smaller than 0.5 mm^2 that do not touch the border of the field (dark lumens, the vignetted tile corners) are added, giving the footprint Ω. *(iv) Slab.* Per column the first and last masked depths $z_\mathrm{top}(x, y)$, $z_\mathrm{bot}(x, y)$ are smoothed laterally over 50 µm (normalised convolution over the measured columns) and interpolated into the filled holes (nearest measured value); the slab is $Z(x, y) = \{z : \lfloor z_\mathrm{top} \rfloor \le z \le \lceil z_\mathrm{bot} \rceil,\ I(x, y, z) \text{ finite}\}$ for $(x, y) \in \Omega$ and empty otherwise. Dark structures between the two surfaces are therefore part of the tissue. *(v) Projections.* The main image selects the depth per pixel: with the signal in dB above the noise floor, $\tilde{I}(x, y, z) = \max(I(x, y, z) - N(z), 0)$, where $N(z)$ is the median (dB) of the columns without tissue at depth z,

$$P_\mathrm{top}(x, y) = \tfrac{1}{k} \textstyle\sum_{z \in \mathcal{T}_k(x,y)} \tilde{I}(x, y, z) \quad \text{[dB above noise]}, \tag{17}$$

where $\mathcal{T}_k$ are the $k = 5$ (10 µm) depths of $Z(x, y)$ with the largest $\tilde{I}$; the values are ≥ 0, without a dark tail. It follows the brightest tissue layer at every pixel instead of diluting it with dark voxels of the slab (as in a reference implementation of the same group that projects the strongest supported slices). We also save

$$P_\mathrm{mean}(x, y) = 20\log_{10}\left(\frac{1}{|Z(x, y)|} \sum_{z \in Z(x,y)} 10^{I(x,y,z)/20}\right), \qquad P_\mathrm{max}(x, y) = \max_{z \in Z(x,y)} I(x, y, z); \tag{18}$$

all are undefined (NaN) for $Z(x, y) = \emptyset$; P_mean averages amplitudes, as the stitching does. The projections and the slab-thickness map $|Z|\ \Delta z$ are saved as float32 TIFF images calibrated in two ways: the standard TIFF resolution in pixels per centimetre, with x and y taken from their own output axes (read by any TIFF reader), and the ImageJ calibration (unit micron); the description holds the acquisition (patches, focus stack, system, probe), the meaning of the values (dB, NaN = no tissue), the stage position of the first pixel and a display range

(1st–99.5th percentile), and a JSON sidecar holds everything. The 8-bit previews carry the physical pixel size; an additional display-only preview normalises the local contrast, $(P-\mu_\sigma)/s_\sigma$ with NaN-aware Gaussian local mean and standard deviation over $\sigma = 80\,\mu\text{m}$, so that features are equally visible in bright and dim regions. For the removed-signal volume the projection is taken over all finite depths. The projection can be recomputed from a kept float volume with other settings without reconstructing again. To quantify tile-periodic artefacts we fold $10^{P_{\text{mean}}/20}$ onto one 500×500 tile by averaging over all tiles, smooth it with a Gaussian ($\sigma = 5\,\text{px}$) and report the difference between its 95th and 5th percentiles in dB ('tile-periodic modulation').

Tile flat-field (seam) correction

The 108 patches were acquired with no overlap (step = field of view = 1 mm), and the texture of neighbouring patches is uncorrelated across a seam (normalised cross-correlation ≈ 0 for all tested overlaps up to 150 pixels and offsets up to ± 60 pixels, versus ≈ 0.5 between adjacent columns inside a patch), so registration and blending are not possible. The visible seams are caused by a brightness falloff towards the patch edges that is the same in every patch. We model the stitched amplitude as

$$A(x, y, z) = N(z) + G(z, u, v)\, S(x, y, z), \tag{19}$$

with (u, v) the position inside the patch, $N(z)$ the noise floor (median of the columns without tissue; for a field fully covered by tissue, the emptiest depth plane) and S the tissue signal. Over the patches t that are fully covered by tissue,

$$G(z, u, v) = \frac{\sum_t X_t(z, u, v)}{\sum_t m_t(z)}, \qquad X = \max(A - N, 0), \quad m_t(z) = \langle X_t(z, \cdot, \cdot)\rangle, \tag{20}$$

i.e. each patch is normalised by its own level so that differences of the tissue between patches average out. G is smoothed laterally (Gaussian, 8 pixels) and along z with signal weights (so that planes without tissue do not pull it to 1), normalised to mean 1 per plane, set to 1 for planes with less than 5 % of the peak signal and capped at 24 dB. The correction rescales only the signal above the noise floor,

$$A' = A + \max(A - N, 0)\left(G^{-1} - 1\right), \tag{21}$$

so that the background at the patch edges is not amplified. It is applied to the stitched volume in place (streamed one patch row at a time) before the TIFF and the projections are written; the gain map is saved.

GPU implementation

The code is written against an array-module abstraction (NumPy [14] or CuPy [15]; PyTorch for Apple GPUs), so the same routines run on CPU and GPU. Raw files of one row of tiles (96 tiles × 500 B-scans) are read tile-major with unbuffered `readinto` into preallocated int16 buffers by a pool of 8 threads, with up to 16 batches of 50 B-scans in flight (reduced automatically if free memory is short), so reading overlaps with computation. Each batch (25,000 A-lines) is copied to the device and processed by one fused CUDA kernel (one thread per output spectral sample) that subtracts the apodisation and spectral means, applies the banded sinc5 operator, re-adds the mean and multiplies by the complex window, writing complex64 spectra; a batched cuFFT computes Eq. (4). The curvature correction is a single gather with the precomputed shift map, restricted to the rows that can reach the output, and stitching is two batched GEMMs (cuBLAS) for numerator and denominator into accumulators for the whole row of tiles ($500 \times 35 \times 6000 \times 2$ float32, ≈ 840 MB) that remain in GPU memory. After the last tile of a row the NaN mask and the dB conversion are applied on the device and one array is copied back. On the CPU the fused step is a Numba kernel parallelised over A-lines and the FFT is multithreaded pocketfft [16,17]. Single precision is used throughout; its error relative to double precision ($\approx 10^{-4}$ dB) is ten times below 1 LSB. Timers synchronise the device and are exclusive. Benchmarks used an NVIDIA RTX 3080 Ti Laptop GPU (16 GB), an AMD Ryzen 9 6900HX (8 cores, 16 threads), 14 GB RAM and an external USB SSD ($\approx 950\,\text{MB}\,\text{s}^{-1}$ sequential read).

Validation

Each stage was compared with the unmodified reference functions run on the same raw B-scan in MATLAB R2024a (maximum relative errors 2.4×10^{-12} for the resampled spectrum, 8.5×10^{-13} for the complex scan in double precision; zero mismatches in the curvature validity map), and the complete reconstruction with the reference volume voxel by voxel. For the film removal, synthetic ground truth was generated by adding PSF reflections (Hann window, sub-pixel tilt $0.8\sin(2\pi x/n_x)$ samples, a phase that is laterally coherent within a B-scan and random between B-scans, ±20 % amplitude jitter) to complex Gaussian speckle filtered axially with a 5-tap Hann kernel and to a 30-dB weaker noise floor (4×200 A-lines). Errors are energies in the 17-sample window relative to the tissue. The unit tests of the software use the same generators with a laterally

smooth film amplitude and require a residual below −6 dB relative to the tissue (or a suppression of at least 25 dB) for reflections 10–30 dB above tissue, an empty-background level within ±3 dB, a change below −20 dB for tissue without reflection, that incoherent bright peaks and films weaker than the tissue are left in place, and that the output has no more B-scan-to-B-scan line structure than the tissue itself; further tests recover a known synthetic vignetting with the flat-field and check the calibration of the 2-D outputs. Ground truth from real data (tissue from below the removal window under a real film reflection of a tile without tissue) was used for the coherence cap. For real data, single B-scans of a film-only tile and a tissue tile were processed on the CPU with the production code, and full rows of tiles and the full volume were reconstructed on the GPU with all version-2 steps off and on. The figures and all numbers in this paper are generated by a script from these results.

Data availability

The raw tiled dataset (956 GB) and the reconstructed volumes are available from the corresponding author on reasonable request [repository and accession to be added]. The intermediate results used for the figures (single-tile complex B-scans, the reconstructed row of tiles with and without removal, en-face projections and timing records) are provided with the code.

Code availability

Version 1 of the reconstruction software (GPU/CPU port of the myOCT tiled-scan reconstruction, web application, validation scripts) is available at `https://github.com/atesfet/oct_recon_AF_v1`. Version 2 (film-reflection removal, en-face projections, removed-signal volume) will be available at `https://github.com/atesfet/oct_recon_AF_v2` [to be made public]. The script that generates all figures and numbers of this paper is included.

Acknowledgements

[Acknowledgements and funding.]

Author contributions

[Author contributions.]

Competing interests

[Competing interests statement.]

References

[1] Huang, D. et al. Optical coherence tomography. *Science* **254**, 1178–1181 (1991).

[2] Fercher, A. F., Drexler, W., Hitzenberger, C. K. & Lasser, T. Optical coherence tomography—principles and applications. *Rep. Prog. Phys.* **66**, 239–303 (2003).

[3] Wojtkowski, M., Leitgeb, R., Kowalczyk, A., Bajraszewski, T. & Fercher, A. F. In vivo human retinal imaging by Fourier domain optical coherence tomography. *J. Biomed. Opt.* **7**, 457–463 (2002).

[4] Leitgeb, R., Hitzenberger, C. K. & Fercher, A. F. Performance of Fourier domain vs. time domain optical coherence tomography. *Opt. Express* **11**, 889–894 (2003).

[5] de Boer, J. F. et al. Improved signal-to-noise ratio in spectral-domain compared with time-domain optical coherence tomography. *Opt. Lett.* **28**, 2067–2069 (2003).

[6] Choma, M. A., Sarunic, M. V., Yang, C. & Izatt, J. A. Sensitivity advantage of swept source and Fourier domain optical coherence tomography. *Opt. Express* **11**, 2183–2189 (2003).

[7] Yolab. myOCT: MATLAB toolbox for OCT acquisition and processing. GitHub `https://github.com/MyYo/myOCT`.

[8] Watanabe, Y. & Itagaki, T. Real-time display on Fourier domain optical coherence tomography system using a graphics processing unit. *J. Biomed. Opt.* **14**, 060506 (2009).

[9] Van der Jeught, S., Bradu, A. & Podoleanu, A. G. Real-time resampling in Fourier domain optical coherence tomography using a graphics processing unit. *J. Biomed. Opt.* **15**, 030511 (2010).

[10] Zhang, K. & Kang, J. U. Real-time 4D signal processing and visualization using graphics processing unit on a regular nonlinear-k Fourier-domain OCT system. *Opt. Express* **18**, 11772–11784 (2010).

[11] Wojtkowski, M. et al. Ultrahigh-resolution, high-speed, Fourier domain optical coherence tomography and methods for dispersion compensation. *Opt. Express* **12**, 2404–2422 (2004).

[12] Schmitt, J. M., Xiang, S. H. & Yung, K. M. Speckle in optical coherence tomography. *J. Biomed. Opt.* **4**, 95–105 (1999).

[13] Donoho, D. L. De-noising by soft-thresholding. *IEEE Trans. Inf. Theory* **41**, 613–627 (1995).

[14] Harris, C. R. et al. Array programming with NumPy. *Nature* **585**, 357–362 (2020).

[15] Okuta, R., Unno, Y., Nishino, D., Hido, S. & Loomis, C. CuPy: a NumPy-compatible library for NVIDIA GPU

calculations. In *Proc. Workshop on Machine Learning Systems (LearningSys) at NIPS* (2017).

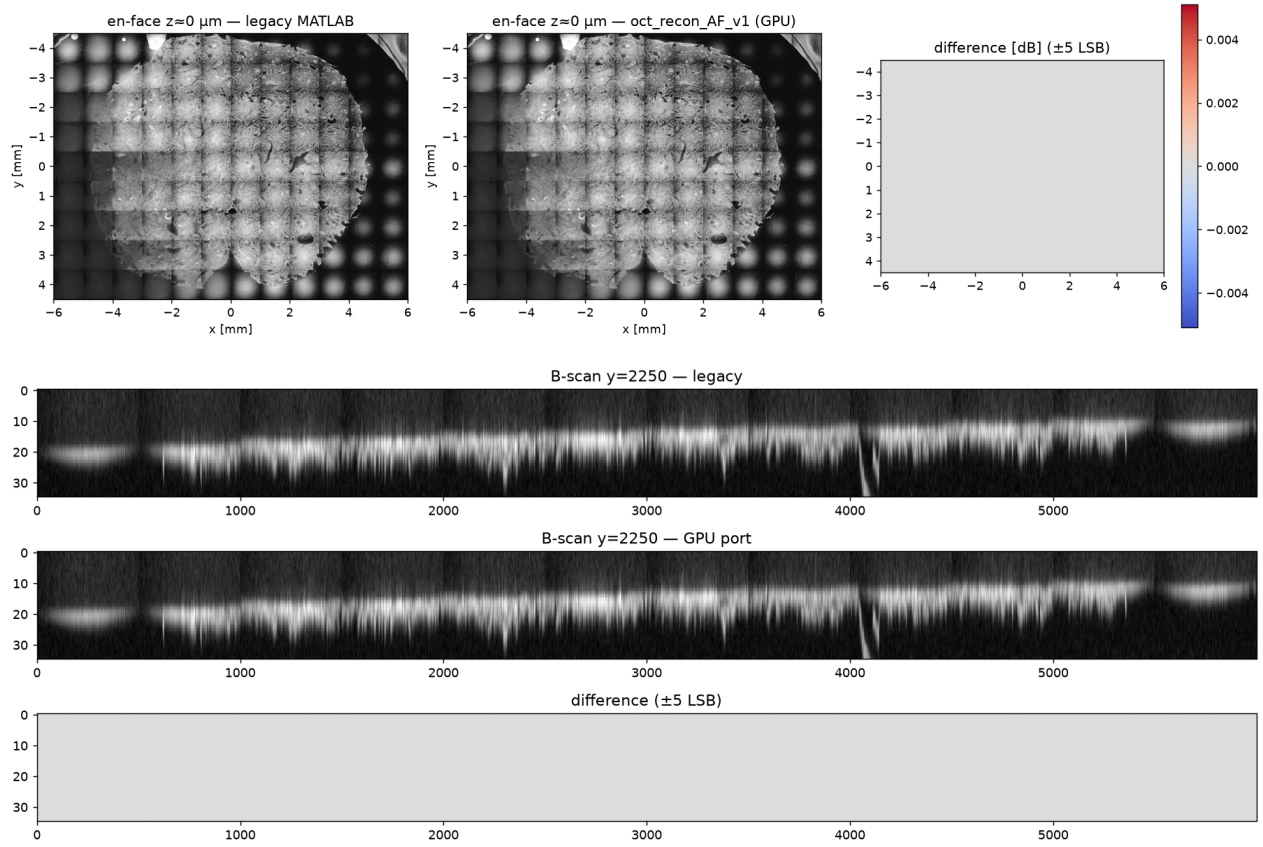
[16] Lam, S. K., Pitrou, A. & Seibert, S. Numba: a LLVM-based Python JIT compiler. In *Proc. Second Workshop on the LLVM Compiler Infrastructure in HPC* 1–6 (2015).

[17] Virtanen, P. et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020).

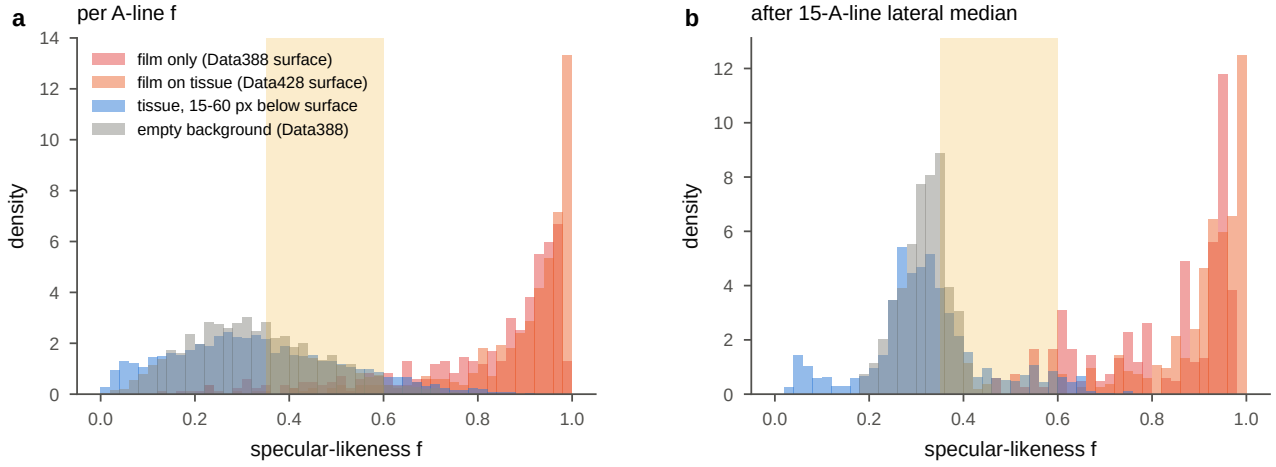
Table 2. Reconstruction and film-removal parameters for the dataset (configuration names of the software in typewriter font).

Parameter	Value	Source / meaning
Spectral pixels N, A-lines, apodisation lines	2048, 500, 25	`Header.xml`
Wavelength range	796–1010 nm	system preset (GAN632)
k-linearisation	sinc5 (`interp_method`)	reference default
Dispersion β	$8.949 \times 10^7\,\mathrm{nm^2\,rad^{-1}}$	probe preset; reproduces reference bit-exactly
Refractive index n	1.33	`ScanInfo.json`
Depth pixel Δz	1.43385 µm (tile), 2 µm (output)	derived; output = lateral pixel size
Optical-path polynomial $p_{1..5}$	0.0099168, 0.010817, −0.0001907, −0.00026705, 4.8047×10^{-12}	probe (`ScanInfo.json`)
Focus pixel m_f, width σ	433 px, 10 px	focus-calibration file; objective preset
Output depth range	−30 to +40 µm (`crop_z_range_mm`)	range of focal depths
Batch, reader threads, prefetch	50 B-scans, 8, 16 batches	execution
Precision	float32 / complex64	execution
FEP removal	on (`fep_removal`)	off = version-1 output
Basis rank r, half-window R	3, 8 (`fep_rank`, `fep_half_window`)	17-sample window
Peak search S, lateral median	±4, 15 A-lines (`fep_search`, `fep_lateral_median`)	
Weight ramp f_lo, f_hi	0.35, 0.60 (`fep_frac_lo/hi`)	specular-likeness
Detection threshold, max. surfaces	6 dB, 4 (`fep_detect_min_db`, `fep_max_surfaces`)	above 51-row median
Background rows B, level κ	10, 0.7 (`fep_keep_level`)	$\kappa = 0$: full projection
Basis	learnt, 12 tiles, ≤ 6000 segments (`fep_learn_basis`)	fallback: theoretical
Coherence cap: lag L, window W, margin μ	on, 4, 31 A-lines, 1.0 (`fep_coh_*`)	demodulated for curvature
Smoothing across B-scans; removed amount	5 B-scans (`fep_y_smooth`); median 5×31	no line artefacts
Tile flat-field	on (`flatfield_correction`); 8 px, cap 24 dB	off = version-1 output
Tissue mask	30 µm smoothing, Otsu, holes $< 0.5\,\mathrm{mm^2}$ filled	`tissue_*`
Projection	top-k, $k = 5$; also slab mean and maximum	`xy_projection`
xy projections, removed volume	on (version 2)	off = version-1 output

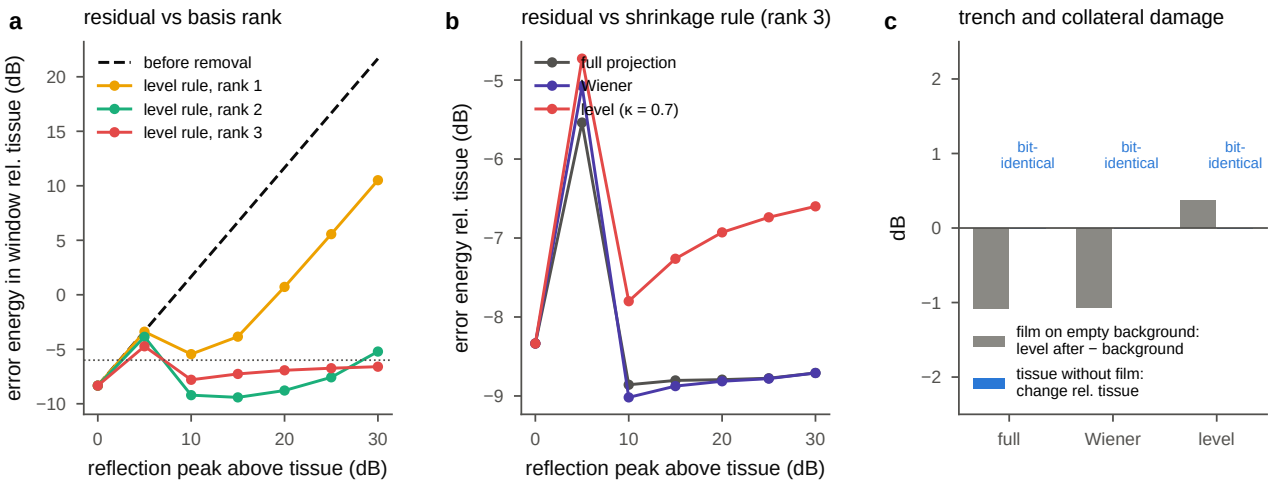
Extended Data



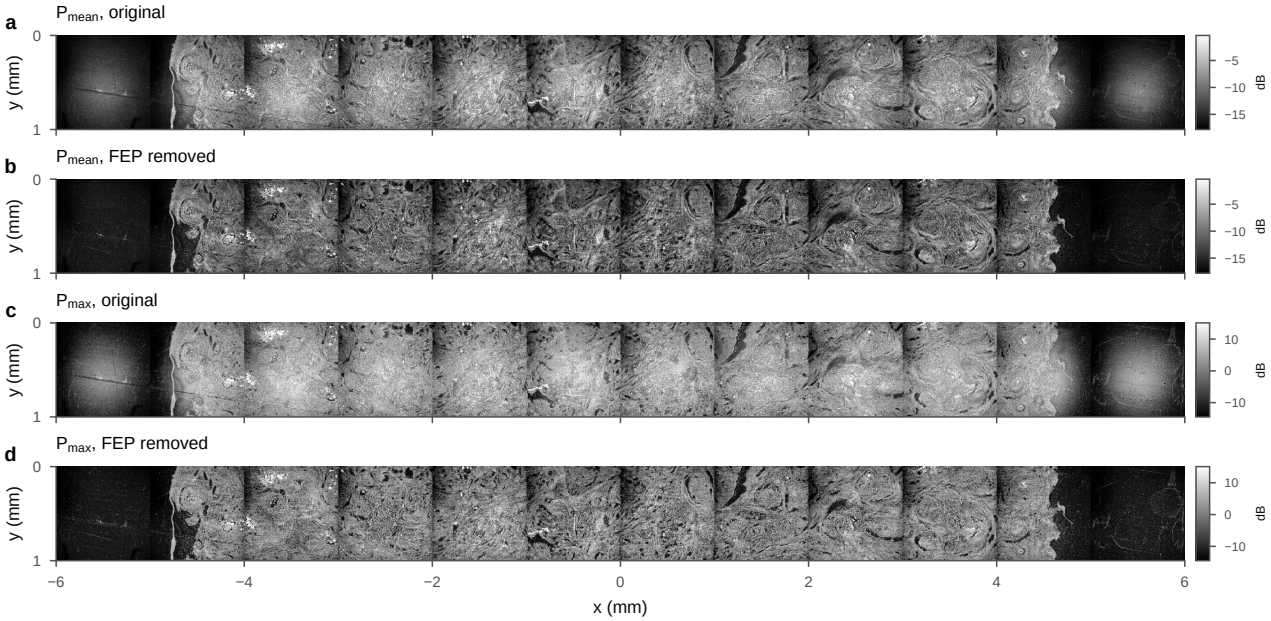
Extended Data Fig. 1. Agreement with the reference reconstruction (version 1, film removal off). Top: en-face plane at $z \approx 0$ over the full 12×9 mm field from the reference MATLAB reconstruction, from the GPU pipeline, and their difference (colour range ± 5 LSB). Bottom: output B-scan (plane 2250) from both and their difference. Over all 945,000,000 voxels the maximum difference is 1 LSB (1.02×10^{-3} dB), 90.7 % of the voxels have identical codes and the masks of missing data are identical; the remaining 1-LSB differences come from the reference's two-step quantisation.



Extended Data Fig. 2. Decision statistic on real data. Specular-likeness $f = \left\|\boldsymbol{U}^{\mathrm{H}}\boldsymbol{s}\right\|^2 / \|\boldsymbol{s}\|^2$ with the learnt rank-3 basis for segments centred on the film surface of a film-only tile and of a tissue tile, for tissue 15–60 samples below the surface and for empty background (one B-scan each, A-lines 40–460). **a**, Per A-line. **b**, After the 15-A-line lateral median used by the method. Yellow, ramp of the weight w. After the median, 95 % of the film positions receive full weight, while 66 % of the tissue and 75 % of the background positions receive zero weight; the remainder receive small weights, which is why processing is additionally restricted to detected surface rows.



Extended Data Fig. 3. Synthetic ground truth. PSF reflections (sub-pixel tilt, laterally coherent phase, ±20 % amplitude jitter; per-B-scan steps of the method) injected into simulated speckle tissue; error energy in the 17-sample window relative to the tissue. **a**, Level rule ($\kappa = 0.7$) with bases of rank 1–3 (theoretical PSF). Dashed, error without removal; dotted, −6 dB acceptance limit of the unit tests. Reflections up to ≈ 5 dB above the tissue fall below the specular-likeness ramp and are not removed. **b**, Rank 3 with the full projection, the Wiener rule and the level rule. The level rule deliberately keeps $\kappa\, r P_{\mathrm{bg}}$ along the subspace, hence a residual about 2 dB higher in tissue. **c**, Reflection 25 dB above an empty noise background: level in the surface rows after removal minus the noise level (grey); tissue without any reflection processed at the same row: the output is bit-identical because the weight w is zero (blue).



Extended Data Fig. 4. En-face projections of one row of tiles (row 4, 12 × 1 mm, all 35 output depths) without and with film-reflection removal, computed from the stitched volumes with the definitions in Methods. **a**,**b**, P_{mean}; **c**,**d**, P_{max}; each pair on the same grey scale. Median P_{mean} −10.2 versus −12.3 dB; median P_{max} 1.8 versus −2.0 dB; tile-periodic modulation of P_{mean} 7.5 versus 4.9 dB.